Climate survey - Japan

OECD

Sample of 1990 respondents.

Results are reweighted along the gender, age, income, highest diploma, region and rural/urban dimensions.

Tables (in .xlsx) corresponding to each figure are available in the /xlsx/JP folder.

September 2022

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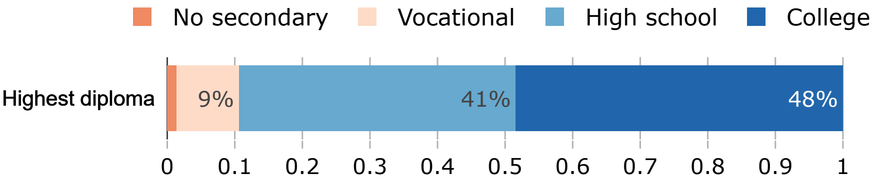
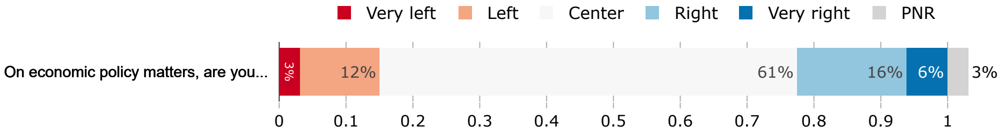
Education

Figure 1: diploma: What is the highest level of education you have completed?



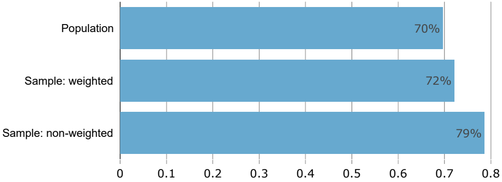
Political affiliation

Figure 2: left_right: On economic policy matters, where do you see yourself on the liberal/conservative spectrum?



Geography

Figure 3: urban: Lives in an urban area (town > 20k people), retrieved from zipcode



Gender and age

Figure 4: gender: What is your gender?

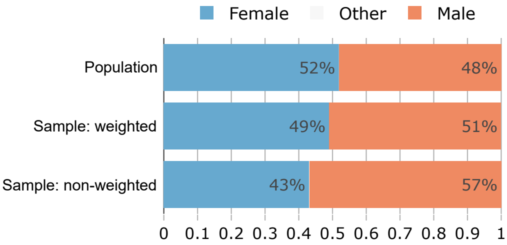
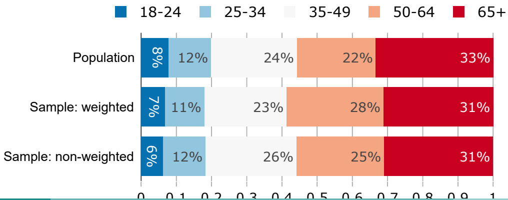


Figure 5: age: How old are you?



Income/wealth

Figure 6: income: What was the annual income of your household in 2019 (before withholding tax, for you and those who live with you)?

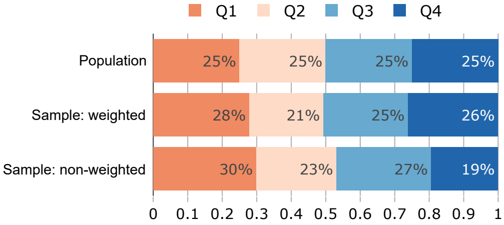
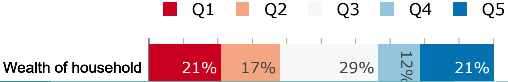


Figure 7: wealth: What is the estimated value of your assets, or the assets of your household if you are married (in Japanese dollars)? Include here all your possessions (home, car, savings, etc.) net of debt. For example, if you own a house worth $300,000 and you have $100,000 left to repay on your mortgage, your assets are $200,000.



Employment and hit by covid

Figure 8: employment_status: What is your employment status?

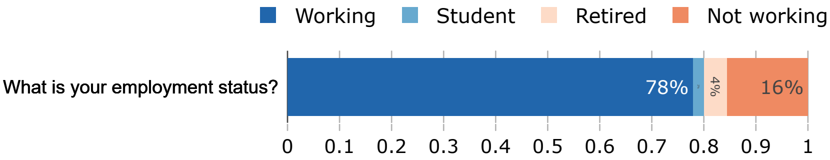
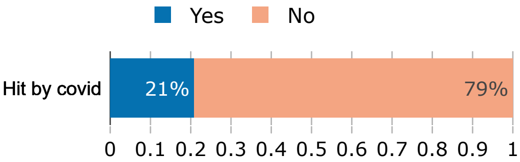


Figure 9: hit_by_covid: Have you or a member of your household been laid off or had to take a cut in your salary or wages due to the COVID-19 pandemic?



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Interest in politics and environmental organizations

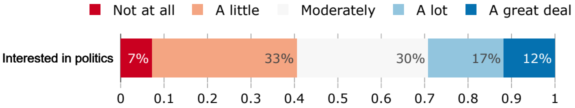


Figure 10: interested_politics: To what extent are you interested in politics?

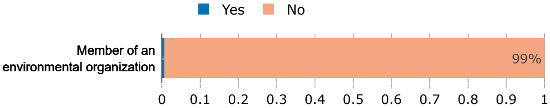


Figure 11: member_environmental_orga: Are you member of an environmental organization?

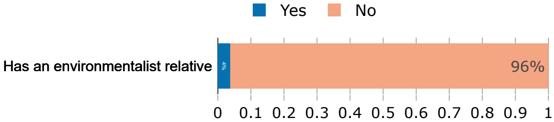
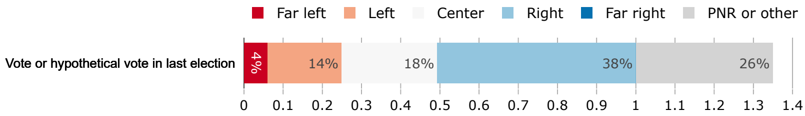


Figure 12: relative_environmentalist: Do you have any relatives who are environmentalists?

Presidential election vote

Figure 13: vote_participation: Did you vote in the 2020 Japanese presidential election?

Figure 14: vote_agg: Which candidate did you vote / would you have voted for in the last presidential election?



Political affiliation

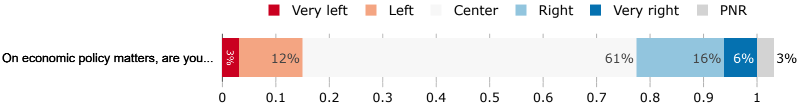


Figure 15: left_right: On economic policy matters, where do you see yourself on the left/right spectrum?

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Figure 16: heating_expenses: In a typical month, how much do you spend on heating for your accommodation?

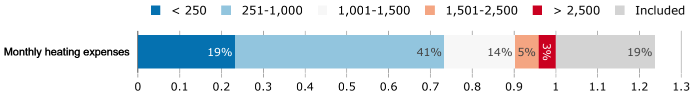


Figure 17: insulation: How do you rate the insulation of your accommodation?

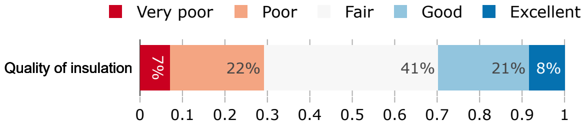


Figure 18: gas_expenses: In a typical month, how much do you spend on gas for driving?

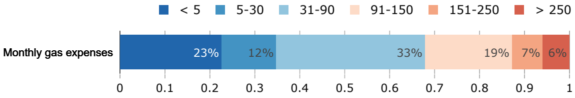


Figure 19: flights_3y: How many round-trip flights did you take between 2017 and 2019?

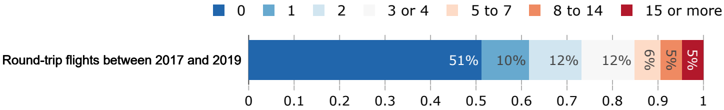


Figure 20: frequency_beef: How often do you eat beef?

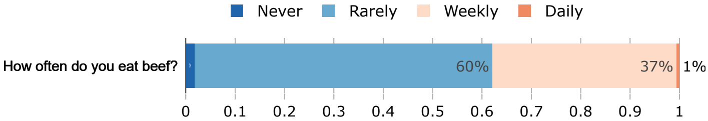


Figure 21: transport: Which mode of transport did you mainly use for each of the following trips in 2019?

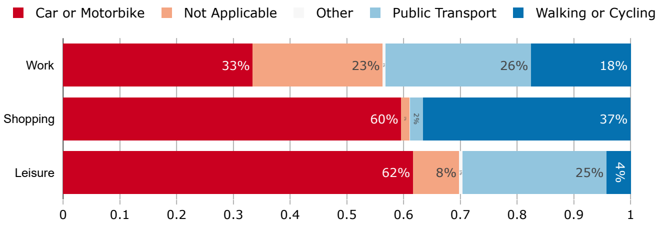
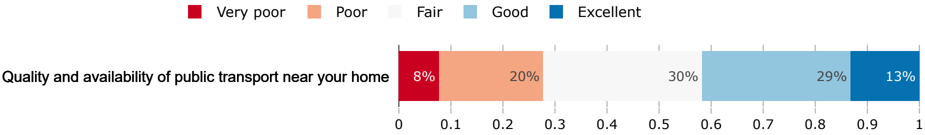


Figure 22: availability_transport: How do you rate the availability (ease of access and frequency) of public transportation where you live?



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Climate change knowledge: general

Figure 23: CC_talks: How often do you think or talk with people about climate change?

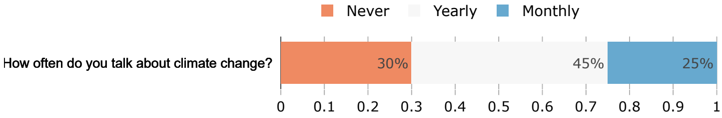
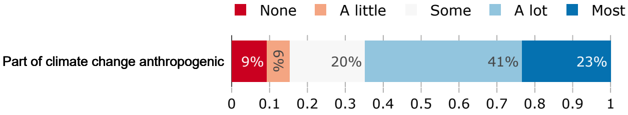


Figure 24: CC_anthropogenic: What part of climate change do you think is due to human activity?



Climate change knowledge: general

Figure 25: CC_knowledgeable: How knowledgeable do you consider yourself about climate change?

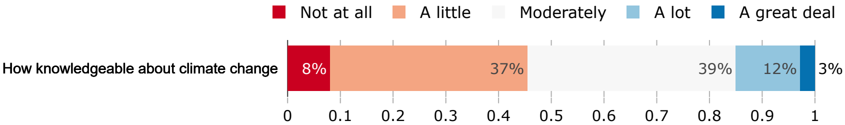


Figure 26: CC_problem: Do you agree or disagree with the following statement: "Climate change is an important problem."

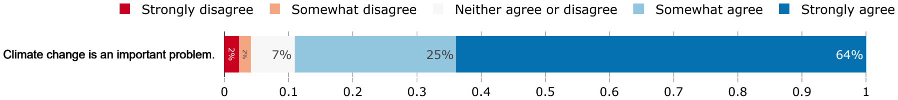
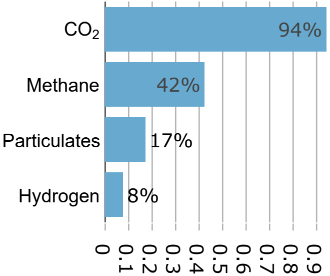
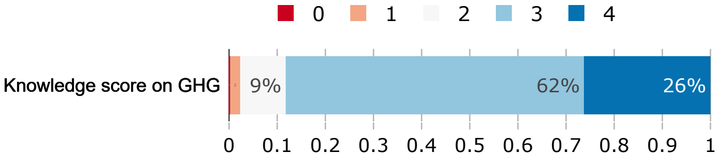


Figure 27: CC_dynamic: Do you think that cutting global greenhouse gas emissions by half would be sufficient to eventually stop temperatures from rising? (Right answer: No)

Climate change knowledge: general

Figure 28: GHG/score_GHG: Which of the following elements contribute to climate change? (Multiple answers are possible)





Score on GHG = CO2 +

Climate change knowledge

Figure 29: footprint_transport: If a family of 4 travels 700 km from Copenhagen to Stockholm, with which mode of transportation do they emit the most greenhouse gases? Please rank the items from 1 (most) to 3 (least).

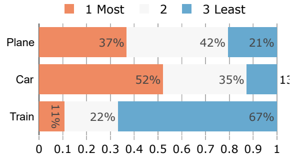
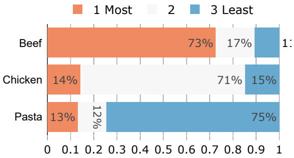
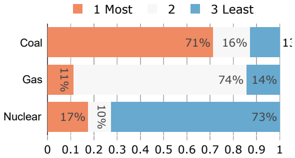


Figure 30: footprint_food: Which dish emits the most greenhouse gases? We consider that each dish weighs half a pound. Please rank the items from 1 (most) to 3 (least).

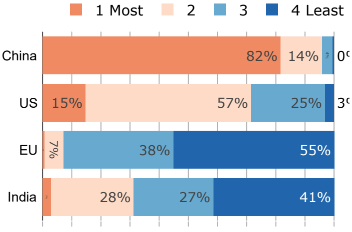


Climate change knowledge

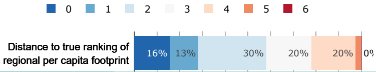
Figure 31: footprint_elec: Which source of electric energy emits the most greenhouse gases to provide power for a house?



(a) footprint_region_no_miss: Which region contributes most to global greenhouse gas emissions?

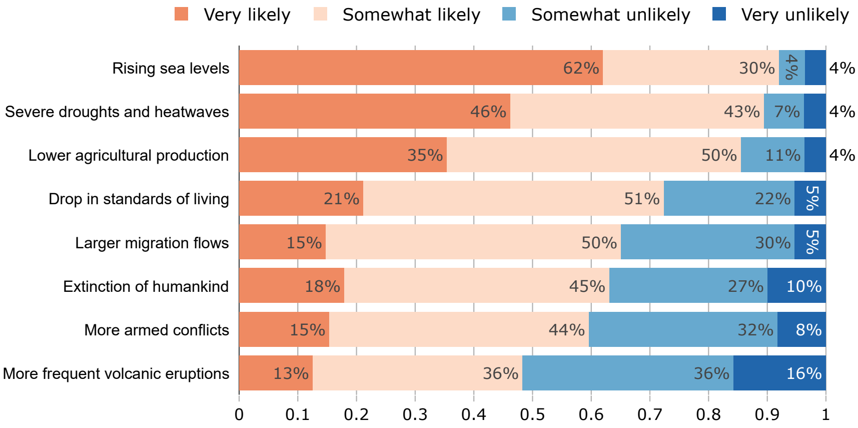


(b) score_footprint_pc: In which region does the consumption of an average person contribute most to climate change? [Kendall distance with true ranking of per capita footprints: US>EU>China>India]



Impacts of climate change

Figure 33: CC_impacts: If nothing is done to limit climate change, how likely do you think it is that climate change will lead to the following events?



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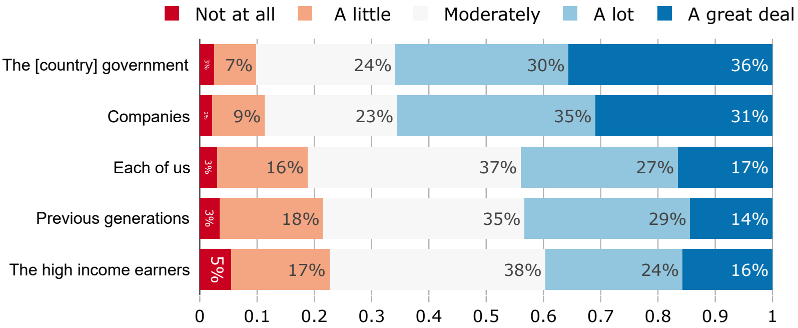
Attitudes and risks

Figure 34: CC_responsible: To what extent are the following groups responsible for climate change in Japan?



Beliefs about the future

Figure 35: net_zero_feasible: To what extent do you think that it is technically feasible to stop greenhouse gas emissions while maintaining satisfactory standards of living in Japan?

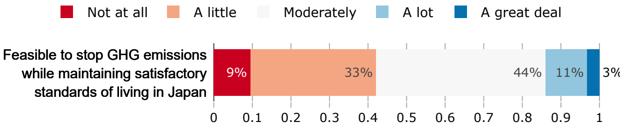
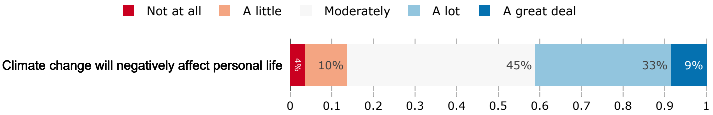


Figure 36: CC_affects_self: To what extent do you think climate change already affects or will negatively affect your personal life?



Beliefs about ambitious climate policies

Figure 37: CC_will_end: How likely is it that human kind halt climate change by the end of the century?

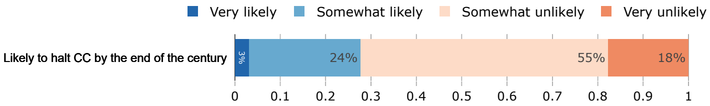


Figure 38: effect_halt_CC_lifestyle: If we decide to halt climate change through ambitious policies, to what extent do you think it would negatively affect your lifestyle?

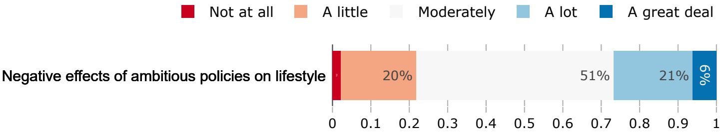
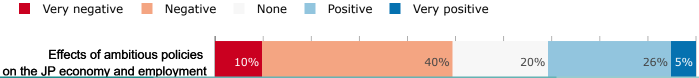
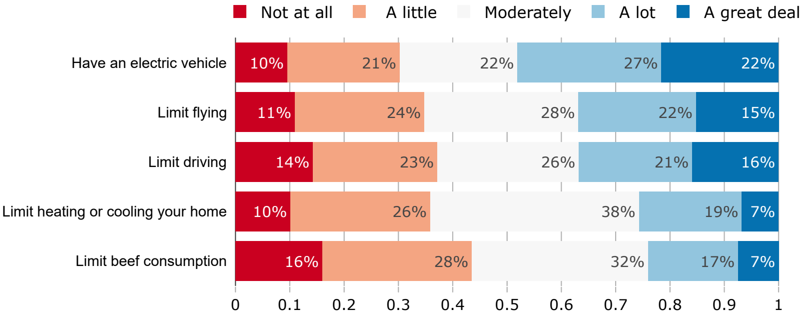


Figure 39: effect_halt_CC_economy: If we decide to halt climate change through ambitious policies, what would be the effects on the U.S economy and employment?



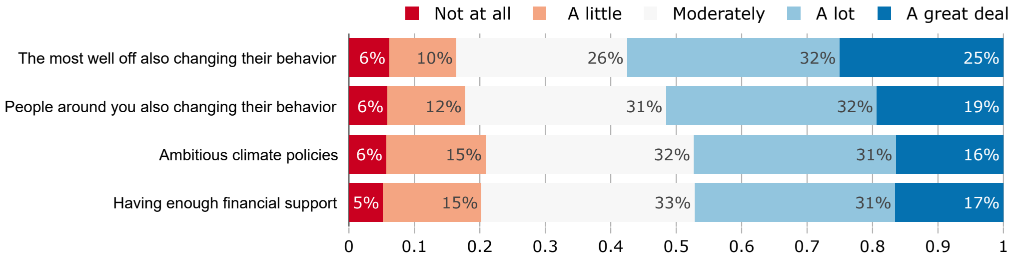
Willingness to change behaviors

Figure 40: willing: Here are possible habits that experts say would help reduce greenhouse gas emissions. To what extent would you be willing to adopt the following behaviors?



Factors needed to change lifestyle

Figure 41: condition: How important are the factors below in order for you to adopt a sustainable lifestyle (i.e. limit driving, flying, and consumption, cycle more, etc.)?



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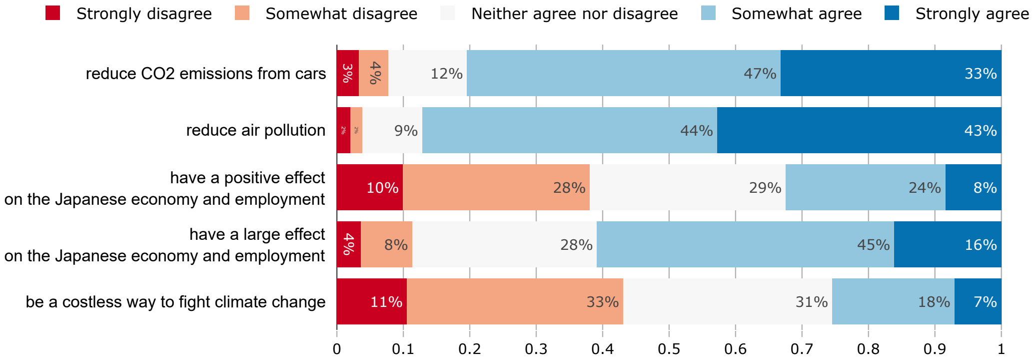
Policy description

To fight climate change, car producers can be required by law to produce cars that emit less CO2 per mile of the cars they sell. The emission limit is lowered every year so that only electric or hydrogen vehicles can be sold after 2030. This policy is called a *ban on combustion-engine cars*.

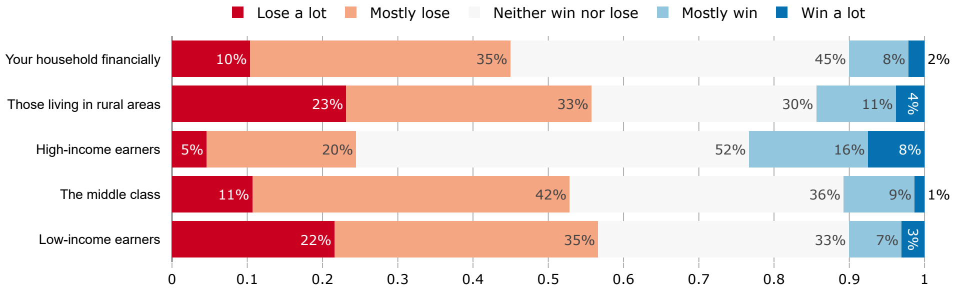
Effects of the policy

Figure 42: standard_effect: Do you agree or disagree with the following statements? A ban on combustion-engine cars would. . .



Incidence

Figure 43: standard_win_lose: In your view, would the following groups win or lose if a ban on combustion-engine cars was implemented in Japan?



Fairness and support

Figure 44: standard_fair: Do you agree or disagree with the following statement: "A ban on combustion-engine cars is fair"?

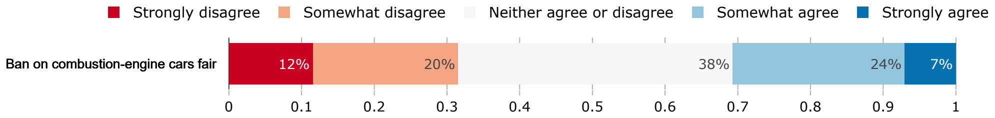


Figure 45: standard_support: Do you support or oppose a ban on combustion-engine cars?

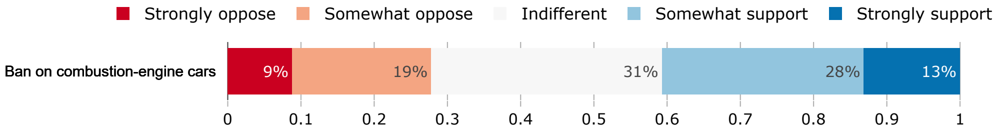
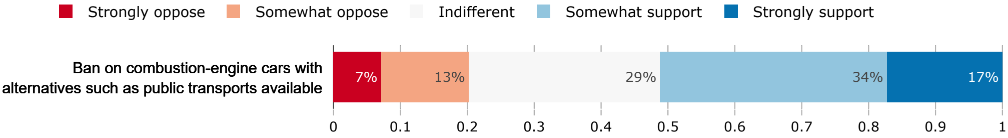


Figure 46: standard_public_transport_support: Do you support or oppose a ban on combustion-engine cars where alternatives such as public transports are made available to people?



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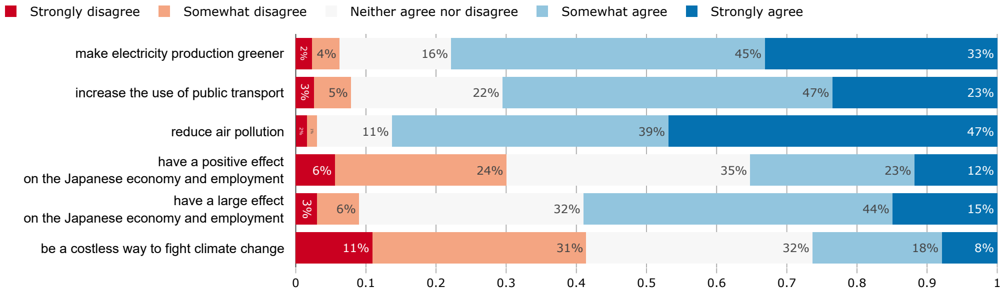
Policy description

A green infrastructure program is a large public investment program, which would be financed by additional public debt, to accomplish the transition needed to cut greenhouse gases emissions. Investments would concern renewable power plants, public transportation, thermal renovation of building, and sustainable agriculture.

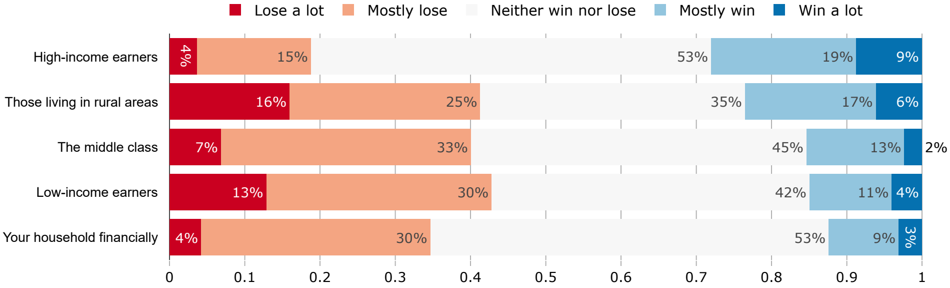
Effects of the policy

Figure 47: investments_effect: Do you agree or disagree with the following statements? A green infrastructure program would. . .



Incidence

Figure 48: investments_win_lose: In your view, would the following groups win or lose with a green infrastructure program?



Fairness and support

Figure 49: investments_fair: Do you agree or disagree with the following statement: “A green infrastructure program mainly financed by public debt is fair."

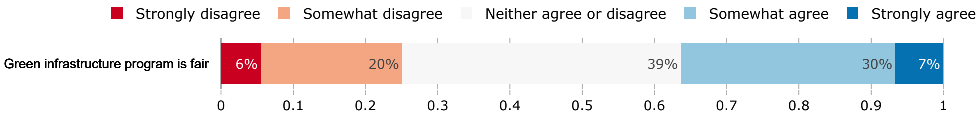


Figure 50: investments_support: Do you support or oppose a green infrastructure program?

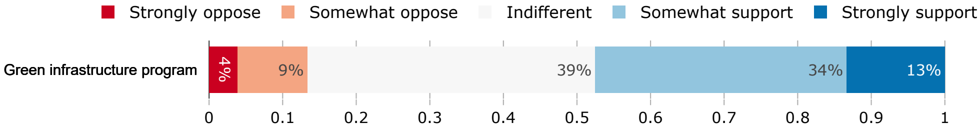
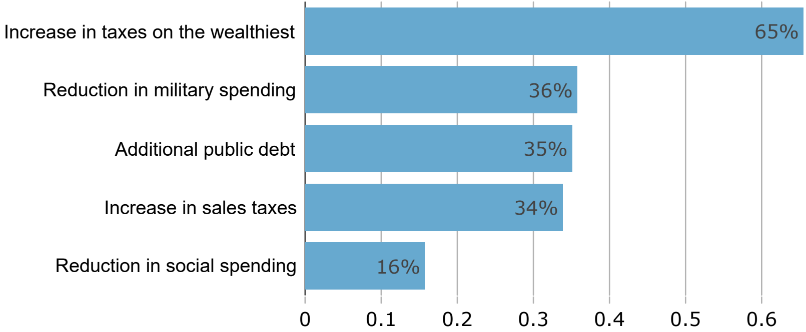


Figure 51: investments_funding: Until now, we have considered that a green infrastructure program would be financed by public debt, but other sources of funding are possible. What sources of funding do you find appropriate for a green infrastructure program? (Multiple answers are possible)



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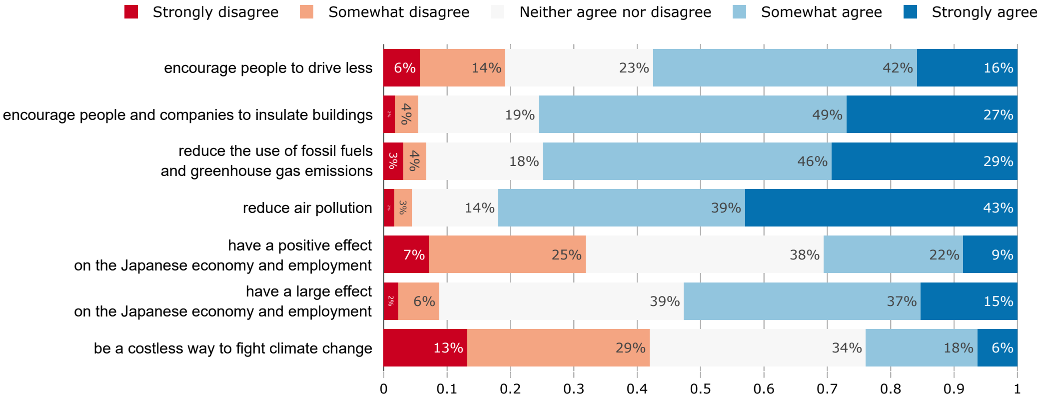
Policy description

To fight climate change, the Japanese government can make greenhouse gas emissions costly, to make people and firms change their equipment and reduce their emissions. The government could do this through a policy called a carbon tax with cash transfers. Under such a policy, the government would tax all products that emit greenhouse gas. For example, the price of gasoline would increase by ¥12/L. To compensate households for the price increases, the revenues from the carbon tax would be redistributed to all households, regardless of their income. Each adult would thus receive [gain_currency] per year.

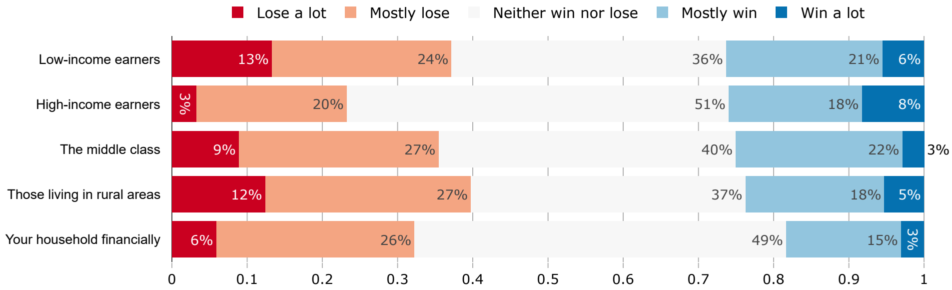
Effects of the policy

Figure 52: tax_transfers_effect: Do you agree or disagree with the following statements? A carbon tax with cash transfers would. . .



Incidence

Figure 53: tax_transfers_win_lose: In your view, would the following groups win or lose under a carbon tax with cash transfers?



Fairness and support

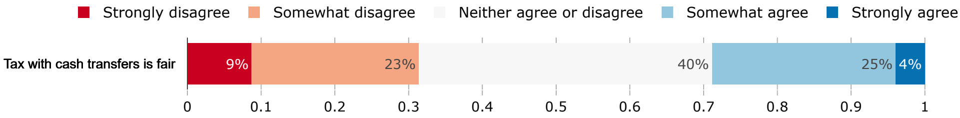


Figure 54: tax_transfers_fair: Do you agree or disagree with the following statement: “A carbon tax with cash transfers is fair."

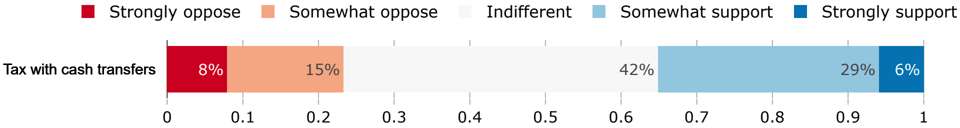


Figure 55: tax_transfers_support: Do you support or oppose a carbon tax with cash transfers?

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Policy Incidence

Figure 56: policies_win_lose_self: *Comparison of responses to each policy question:* Do you think that financially your household would win or lose from *the policy*?

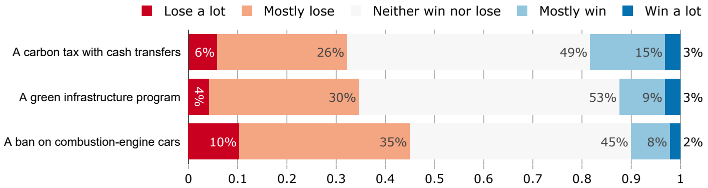
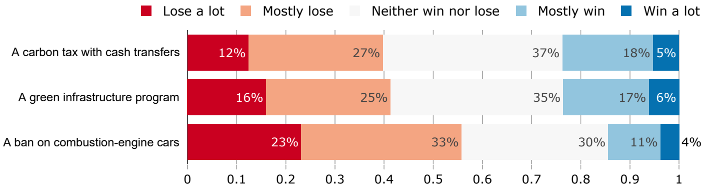


Figure 57: policies_win_lose_rural: *Comparison of responses to each policy question:* In your view, would those living in rural areas win or lose from *the following policy*?



Policy Incidence

Figure 58: policies_win_lose_rich: *Comparison of responses to each policy question:* In your view, would high-income earners win or lose from *the following policy*?

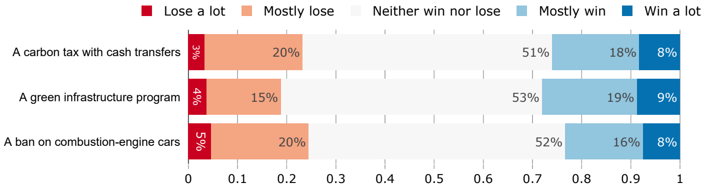
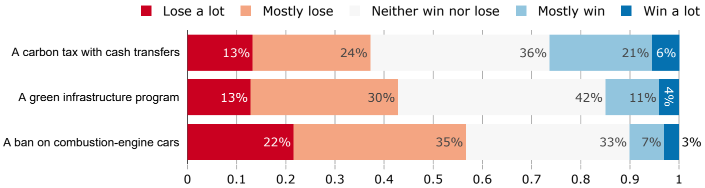
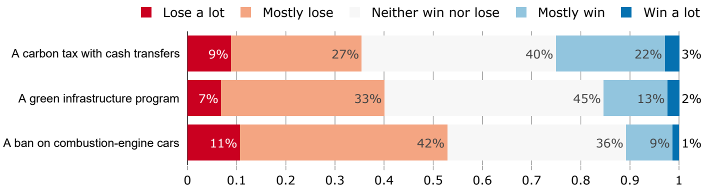


Figure 59: policies_win_lose_poor: *Comparison of responses to each policy question:* In your view, would low-income earners win or lose from *the following policy*?



Policy Incidence

Figure 60: policies_win_lose_middle: *Comparison of responses to each policy question:* In your view, would the middle-class win or lose from *the following policy*?



Effects of the policy

Figure 61: policies_large_effect: *Comparison of responses to each policy question:* Do you agree or disagree with the following statement? *The policy* would have a large effect on the Japanese economy and employment.

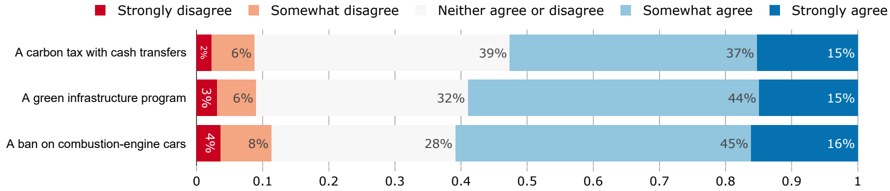
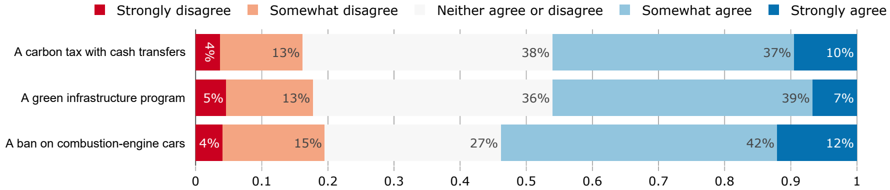
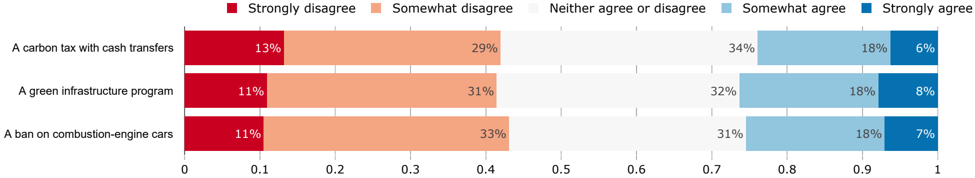


Figure 62: policies_negative_effect: *Comparison of responses to each policy question:* Do you agree or disagree with the following statement? *The policy* would have a negative effect on the Japanese economy and employment.



Effects of the policy

Figure 63: policies_costless_costly: *Comparison of responses to each policy question:* Do you agree or disagree with the following statement? *The policy* would be costless to fight climate change



Fairness and support

Figure 64: policies_fair: *Comparison of responses to each policy question:* Do you agree or disagree with the following statement:"The *policy* is fair."

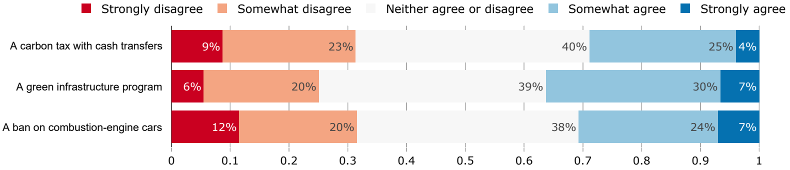
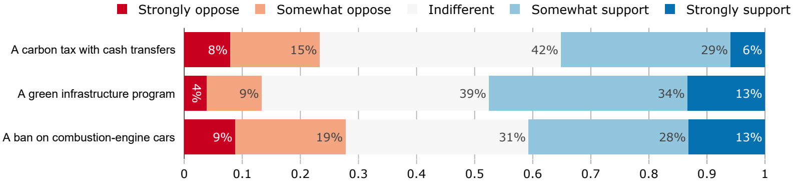


Figure 65: policies_support: *Comparison of responses to each policy question:* do you support or oppose *the following policy*?



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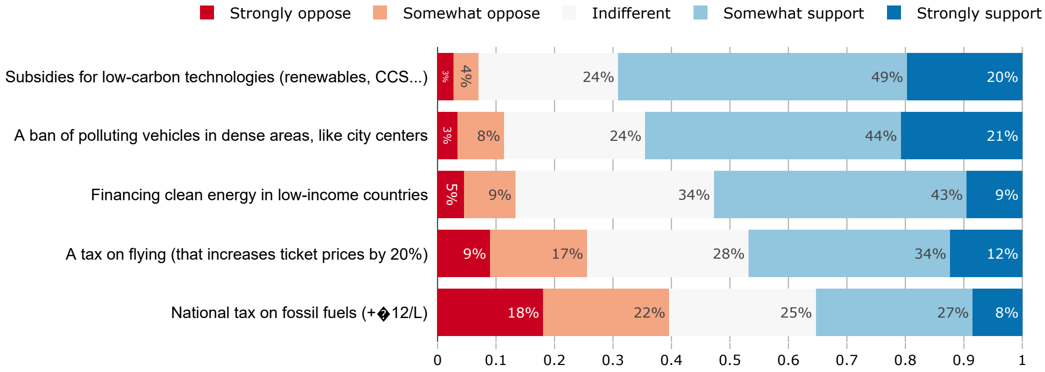
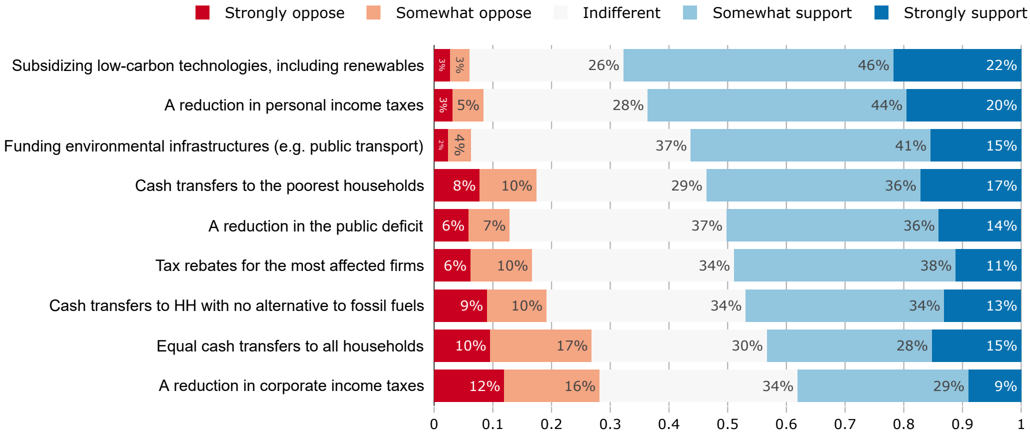
Other policies

Figure 66: policy: Do you support or oppose the following climate policies?



Revenue recycling of carbon tax

Figure 67: tax: Governments can use the revenues from carbon taxes in different ways. Would you support or oppose introducing a carbon tax that would raise gasoline prices by ¥12/L, if the government used this revenue to finance...



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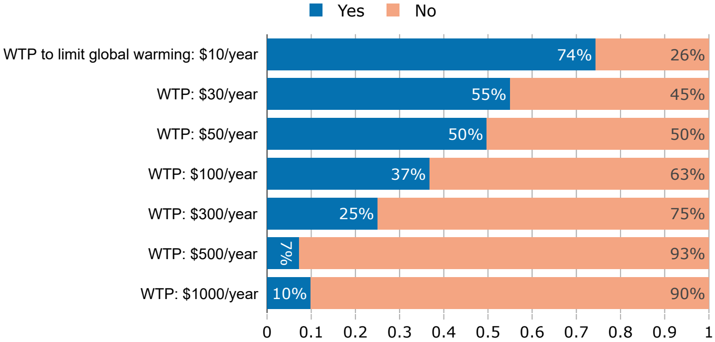
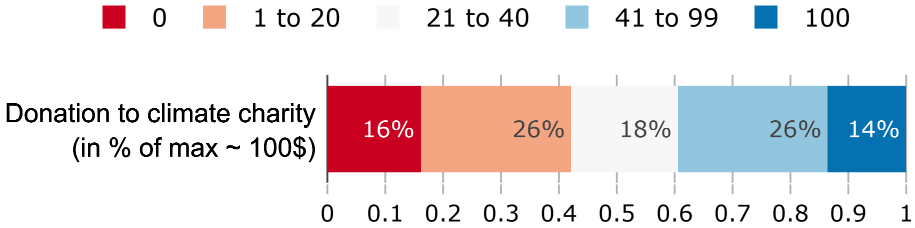
WTP

Figure 68: wtp: To fight global warming, the Japanese government could implement a policy package to reduce emissions, for example by investing in clean technologies (renewable energy, electric vehicles, public transport, more efficient insulation, etc.).
The funding for these investments could be collected annually through an additional individual contribution for the foreseeable future. Assume that everyone in Japan as well as citizens of other countries would be required to contribute according to their means.
Are you willing to pay [amount] annually through an additional individual contribution to limit global warming to safe levels (less than 2 °C)?



Donation

Figure 69: donation_agg: By taking this survey, you are entered into a lottery to win [$\tilde{\$}$100] (100$). You can also donate a part of this additional compensation (should you be selected in the lottery) to a reforestation project through the charity The Gold Standard. If you win the [$\tilde{\$}$100] lottery, how much will you donate to the Gold Standard charity?



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Insulation

Figure 70: will_insulate: How likely is it that you will improve the insulation or replace the heating system of your accommodation over the next 5 years?

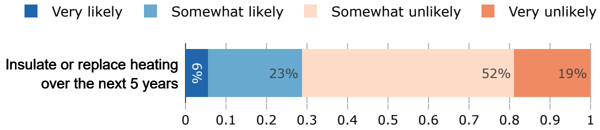
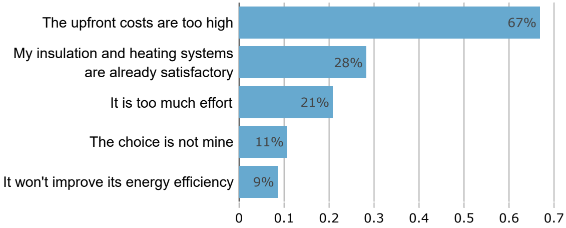
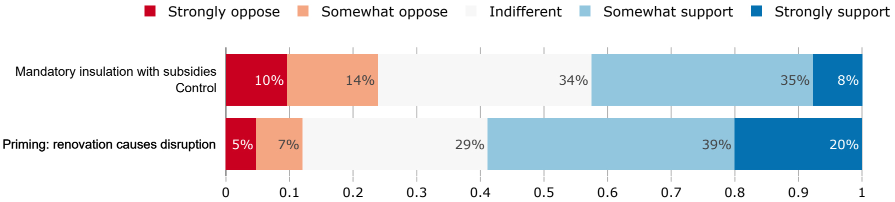


Figure 71: obstacles_insulation: What are the main hurdles preventing you from improving the insulation or replace the heating system of your accommodation? (Multiple answers are possible)



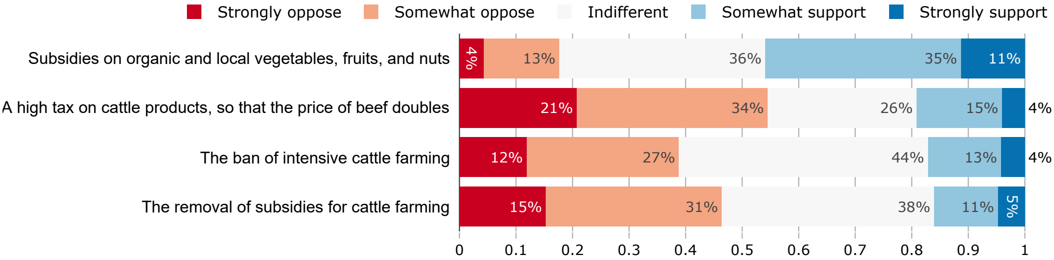
Insulation

Figure 72: insulation_support_variant: Imagine that the Japanese government makes it mandatory for all residential buildings to have insulation that meets a certain energy efficiency standard before 2040. The government would subsidize half of the insulation costs to help households with the transition.
Displayed in disruption variant: [Insulating your home can take long, may cause disruptions to your daily life during the renovation works, and may even require you to leave your home until the renovation is completed.]
Do you support or oppose such policy?



Cattle products

Figure 73: beef: Imagine that, in order to fight climate change, the Japanese government decides to limit the consumption of cattle products like beef and dairy. Do you support or oppose the following options?



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Trust

Figure 74: can_trust_people: Do you agree or disagree with the following statement: "Most people can be trusted."

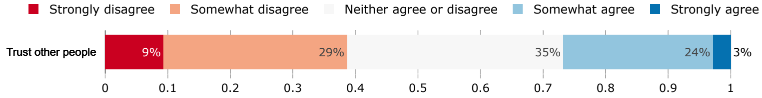
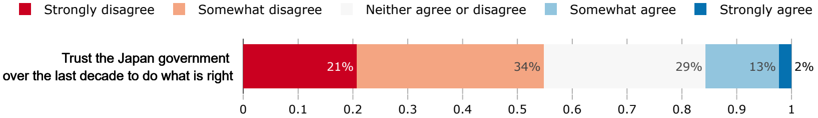


Figure 75: can_trust_govt: Do you agree or disagree with the following statement: "Over the last decade the Japanese government could generally be trusted to do what is right."



Perception of institutions, inequality, and the future

Figure 76: view_govt: Some people think the government is trying to do too many things that should be left to individuals and businesses. Others think that government should do more to solve our country's problems. Which come closer to your own view?

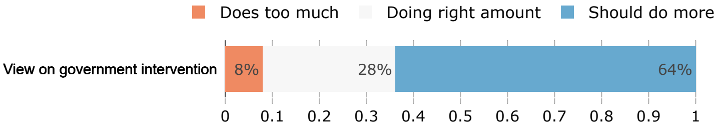


Figure 77: problem_inequality: How big of an issue do you think income inequality is in Japan?

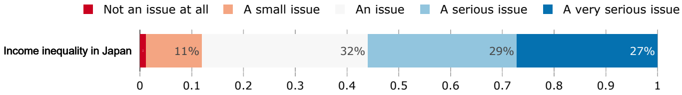
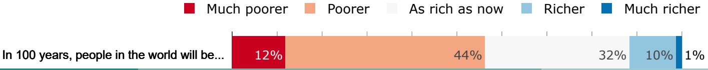


Figure 78: future_richness: Do you think that overall people in the world will be richer or poorer in 100 years from now?



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Willingness to change behavior

Figure 79: willing_limit_flying: To what extent would you be willing to adopt the following behaviors? -– Limit Flying, by Income

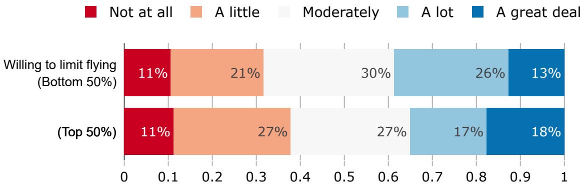
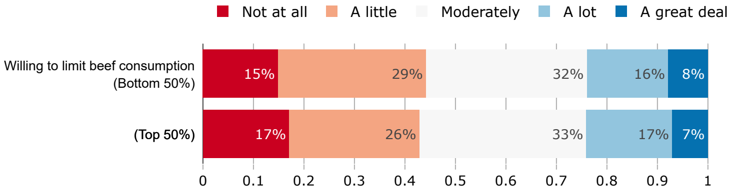


Figure 80: willing_limit_beef: To what extent would you be willing to adopt the following behaviors? – Limit Beef Consumption, by Income



Perception

Figure 81: future_richness: Do you think that overall people in the world will be richer or poorer in 100 years from now? -– by Income

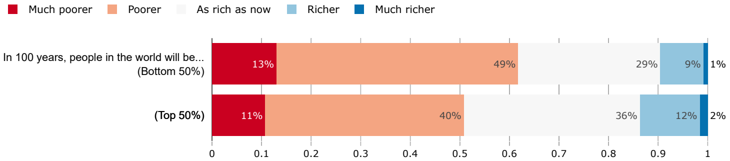
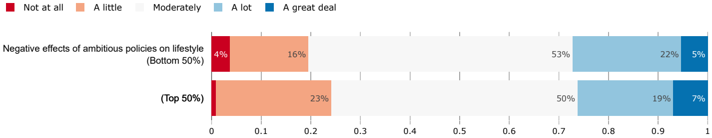


Figure 82: effect_halt_CC_lifestyle: If we decide to halt climate change through ambitious policies, to what extent do you think it would negatively affect your lifestyle? – by Income



Effects on own household

Figure 83: policies_win_lose_self: Do you think that financially your household would win or lose from the following policy? – by Income

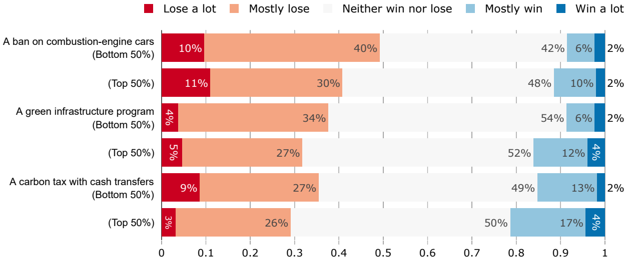
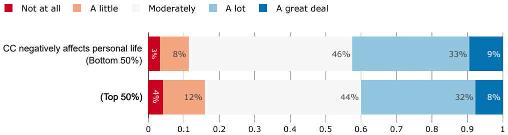


Figure 84: CC_affects_self: To what extent do you think climate change already affects or will negatively affect your personal life? – by Income



Policies – support

Figure 85: policies_support: Do you support or oppose the following policy? – by Income

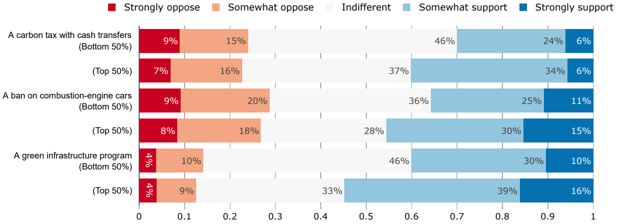


Figure 86: global_assembly_support: Do you support or oppose establishing a global democratic assembly whose role would be to draft international treaties against climate change? Each adult across the world would have one vote to elect members of the assembly. – by Income

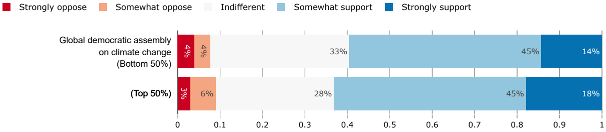


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Table 1: Attitudes towards Climate Change

	CC caused by humans	CC likely to cause extinction	Donation (in % of max)	JP should fight CC	A lot willing to limit driving
	(1)	(2)	(3)	(4)	(5)
Control group mean	0.648	0.628	3565.924	0.429	0.368
Treatment: Climate	0.087^{***}	0.040	249.935	0.078^{**}	0.004
	(0.033)	(0.034)	(232.853)	(0.034)	(0.035)
Treatment: Policy	−0.009	−0.006	−29.986	0.051	−0.049
	(0.035)	(0.036)	(235.950)	(0.035)	(0.035)
Treatment: Both	0.047	−0.003	−7.745	0.019	0.001
	(0.034)	(0.035)	(233.656)	(0.034)	(0.035)
Observations	1,989	1,990	1,990	1,990	1,990
R^2	0.062	0.028	0.099	0.112	0.024

Note: The *CC caused by humans* indicator variable equals one if the respondent thinks a lot or most of climate change is due to human actions. The *CC likely to cause extinction* indicator variable equals one if the respondent thinks climate change is somewhat likely or very likely to cause the extinction of humankind if nothing is done to limit it. The *Donation* variable is a continuous variable equal to the amount the respondent is willing to give to a charity. The *Ambitious policies needed* indicator variable equals one if the respondent thinks policy must be a lot or a great deal ambitious in order to halt climate change. The *Willing to limit driving* indicator variable equals one if the respondent is willing a lot or a great deal to limit driving. The three *treatment* indicator variables indicate difference in mean compared to the control group (people who did not see any video). Controls include socio-demographic, economic affiliation, last vote and whether the respondent's household was hit by the COVID-19 pandemic. Standard errors are in parentheses.

*$p<0.1$; **$p<0.05$; ***$p<0.01$

Table 2: Support for policies

	Support			
	Carbon tax with transfers	Green Infrastructure Program	Ban on combustion-engine cars	Average over 3 policies
	(1)	(2)	(3)	(4)
Control group mean	0.351	0.475	0.407	0.523
Treatment: Climate	0.027	0.049	0.024	0.037
	(0.034)	(0.036)	(0.035)	(0.036)
Treatment: Policy	0.117***	0.082**	0.090**	0.067*
	(0.036)	(0.036)	(0.036)	(0.036)
Treatment: Both	0.147***	0.072**	0.097***	0.082**
	(0.035)	(0.035)	(0.034)	(0.035)
Observations	1,990	1,990	1,990	1,990
R^2	0.058	0.075	0.079	0.074

Note: The dependent variables are indicator variables equal to one if the respondent 'Strongly supports" or "Somewhat supports" the policy. The *Average over 3 policies* takes the average of the respondent's answers for the three policies. It equals one if the respondent support all three policies, $2/3$ if she supports two, $1/3$ if she support only one, and 0 if she supports none. See notes under previous Table for a description of the covariates.

Controls include socio-demographic, economic affiliation, last vote and whether the respondent's household was hit by the COVID-19 pandemic. Standard errors are in parentheses. *p$<$0.1; **p$<$0.05; ***p$<$0.01

Table 3: Attitudes towards policies

	Fair	HH would win	Poor would win	Large economic effect	Positive economic effect
	(1)	(2)	(3)	(4)	(5)
Control group mean	0.396	0.168	0.195	0.725	0.395
Treatment: Climate	0.054	0.043	0.089^{***}	-0.026	0.021
	(0.036)	(0.029)	(0.031)	(0.033)	(0.036)
Treatment: Policy	0.072^{**}	0.104^{***}	0.231^{***}	0.014	0.009
	(0.037)	(0.030)	(0.033)	(0.033)	(0.037)
Treatment: Both	0.114^{***}	0.097^{***}	0.274^{***}	0.034	0.011
	(0.035)	(0.029)	(0.032)	(0.031)	(0.035)
Observations	1,990	1,972	1,971	1,990	1,990
R^2	0.050	0.021	0.081	0.029	0.020

Note: The dependent variables are discrete variables equal either to 0, $1/3$, $2/3$, or 1. They are equal to the average over the three policies mentioned in Table "Support policies". The *Fair* variable equals one if the respondent strongly agrees or somewhat agrees that each of the three policies are fair. The *HH/Poor would win* variables equal one if the respondent thinks her househould/the poorest would win a lot or mostly win from the three policies. The *Large/Negative economic effect* variables equal one if the respondent strongly agrees or somewhat agrees that the three policies would have a large/negative impact on the Japanese economy and employment.

Controls include socio-demographic, economic affiliation, last vote and whether the respondent's household was hit by the COVID-19 pandemic.

Standard errors are in parentheses. *$p<0.1$; **$p<0.05$; ***$p<0.01$